

MULTI-MODAL RIGIDITY SENSING ARCHITECTURE

General Rigidity Detection Concept

Penile rigidity and erection state may be inferred through measurement of mechanical deformation, tissue interaction forces, dynamic stiffness response, and physiologic motion behavior. Rather than relying on a single sensing mechanism, the disclosed device may employ a multi-modal sensing architecture in which multiple sensor types independently or cooperatively contribute to determination of physiologic state. Rigidity may therefore represent a derived parameter based on correlated sensor outputs rather than direct measurement of any single physical property.

Sensor modalities may operate individually, redundantly, or in weighted computational combinations. Different embodiments may include simplified sensing configurations optimized for cost efficiency or expanded sensing architectures designed for higher-fidelity physiologic characterization.

Strain-Based Measurement Embodiments

Strain sensing elements may be incorporated into structural regions of the wearable device to detect circumferential expansion, elastic deformation, or structural displacement associated with tumescence. Strain sensors may include metallic foil strain gauges, printed conductive traces, piezoresistive materials, capacitive strain structures, conductive elastomer sensors, fiber-based strain elements, or other deformation-sensitive technologies. Placement may occur along inner or outer device surfaces or within structural members of open-gap configurations.

Pressure and Force Measurement Embodiments

Pressure sensing elements may measure localized contact forces between the wearable and tissue. Such sensors may include force-sensitive resistors, capacitive pressure sensors, piezoresistive pressure devices, microfluidic pressure channels coupled to transducers, distributed pressure arrays, or compliant force detection structures. Pressure measurements may contribute to rigidity indices, fit stability determination, and assessment of mechanical response during erection onset and detumescence.

Piezoelectric Dynamic Response Sensing

Piezoelectric elements may detect transient mechanical events including micro-vibrations, pulsatile responses, or rapid deformation changes. Such sensors may provide temporal information regarding dynamic stiffness behavior, physiologic pulsation signatures, or event detection useful for episode segmentation and signal validation.

Magnetic Gap and Displacement Measurement

In embodiments incorporating an open-gap configuration, a magnetic element may be positioned on one arm of the device while a Hall effect sensor or equivalent magnetic field detector is positioned on an opposing arm. Variation in magnetic field strength resulting from gap displacement may allow inference of expansion, compression, stiffness response, or rigidity level. Multiple magnets, sensor arrays, shielding structures, or alternative magnetic geometries may be employed to improve linearity and measurement range.

Sensor Substitution and Hybrid Architectures

Any sensing modality described herein may be substituted with alternative technologies capable of detecting substantially similar mechanical or physiologic changes. Hybrid architectures combining strain, pressure, piezoelectric, magnetic, capacitive, optical, acoustic, impedance-based, or future sensing technologies are considered within the scope of this disclosure. Sensor fusion algorithms may integrate signals to produce derived rigidity metrics resilient to noise, motion artifacts, or individual anatomical variability.

The embodiments described are illustrative and non-limiting, and the invention encompasses present and future sensing technologies capable of enabling longitudinal rigidity characterization through multi-modal physiologic inference.